HKE SOCIETY'S

POOJYA DODDAPPA APPA COLLEGE OF ENGINEERING

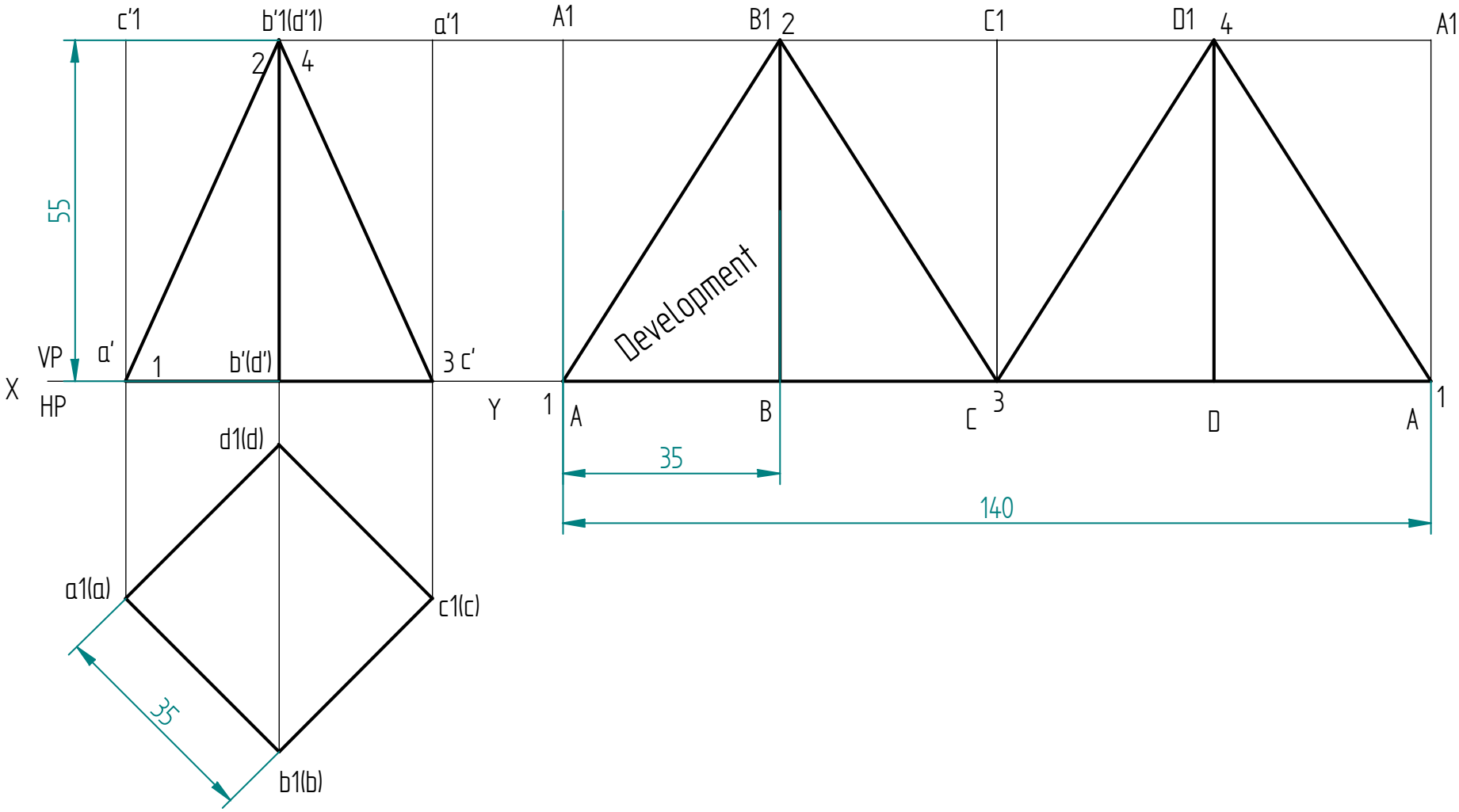
COMPUTER AIDED ENGINEERING DRAWING

I / II SEMESTER

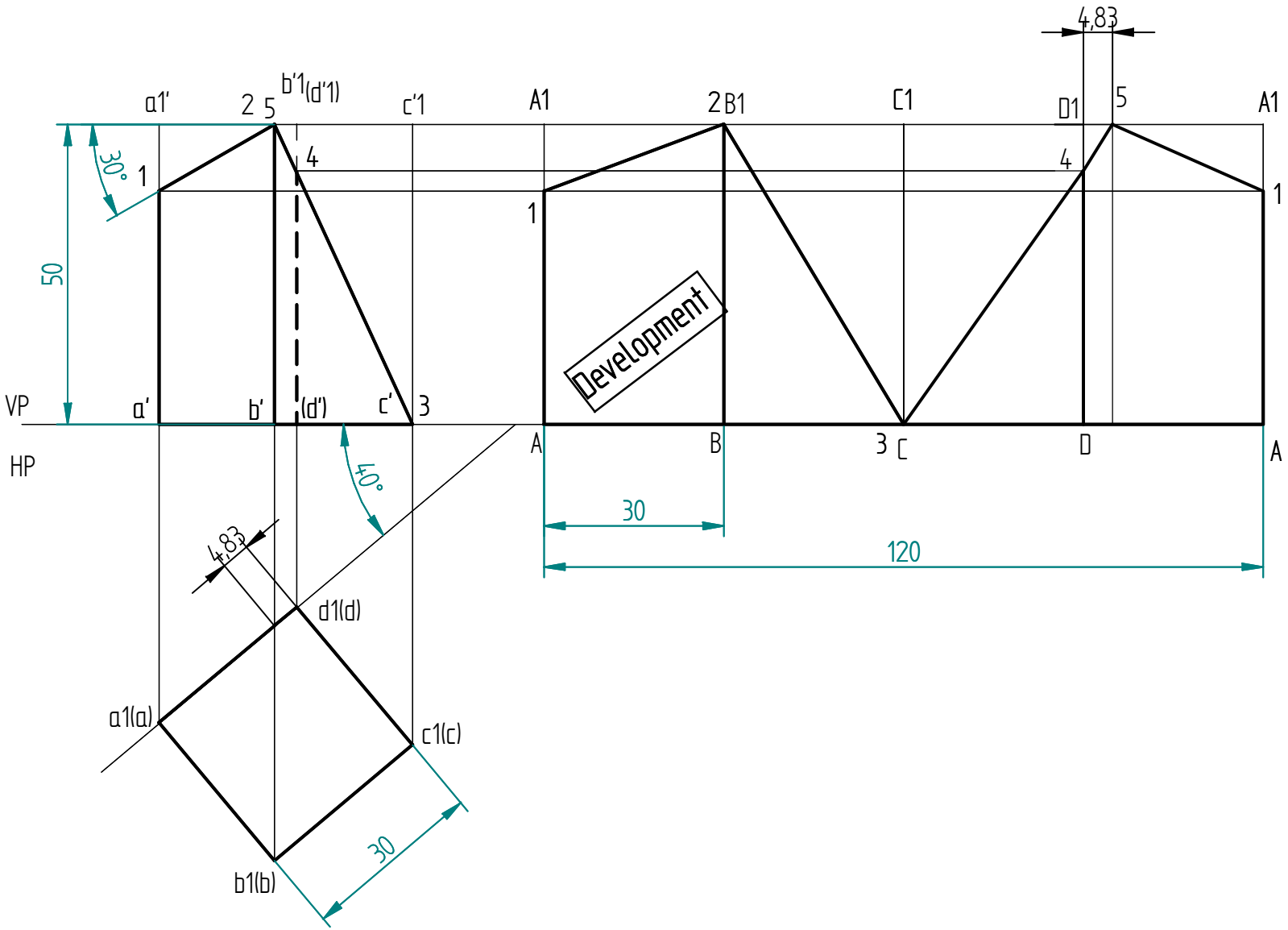
Solutions(Developments)

POOJYA DODDAPPA APPA COLLEGE OF ENGINEERING,
(AUTONOMUS INSTITUTION, AFFILIATED TO VTU BELGAUM)
Aiwan-E-Shahi Area, KALABURAGI-585 102

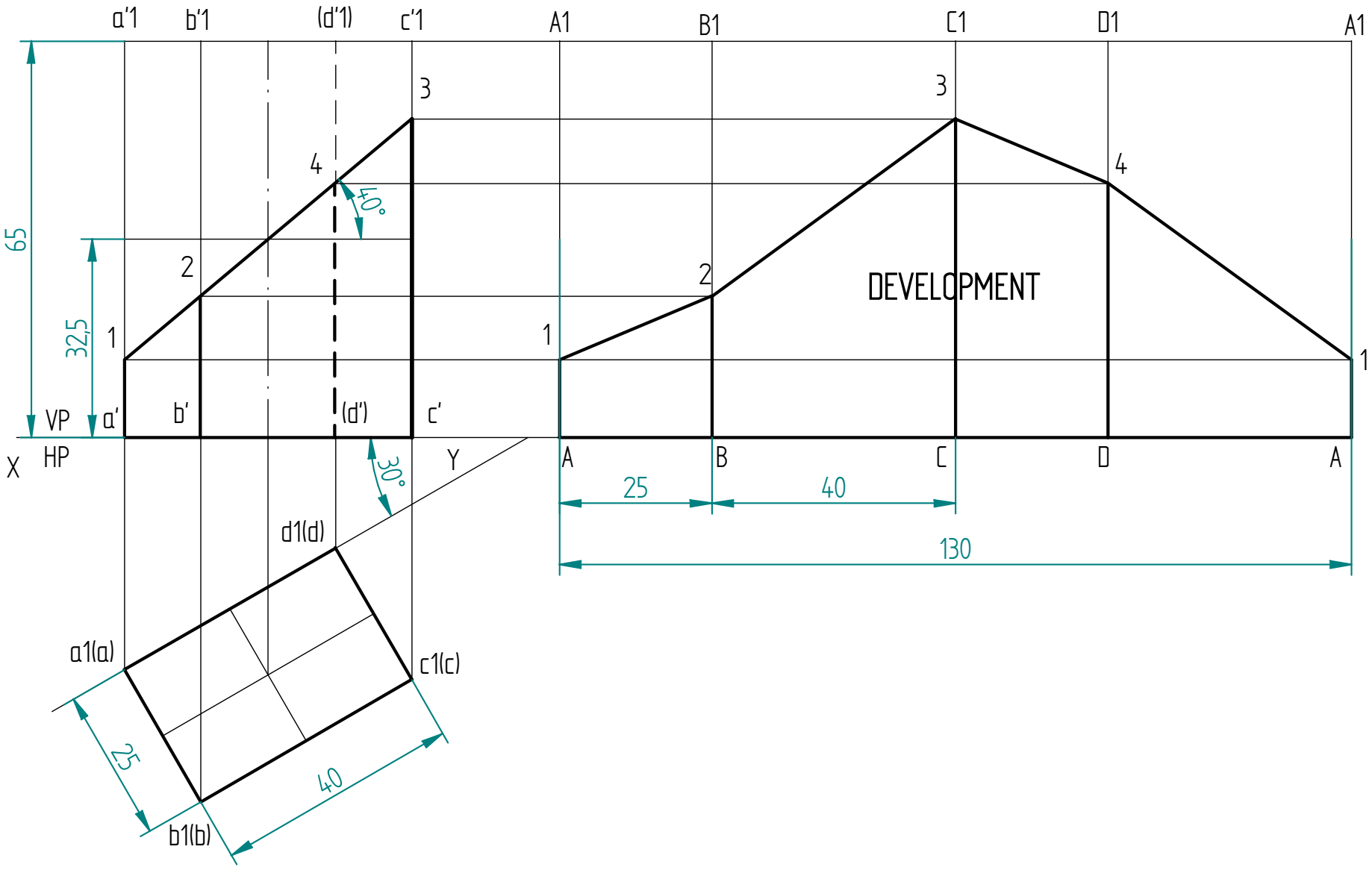
4.1 A Square prism of base side 35mm & height 55mm rests with its base on HP and two faces equally inclined to VP. Draw the development of lateral surfaces of the retained portions of the cut prism shown by dark lines in the figure.



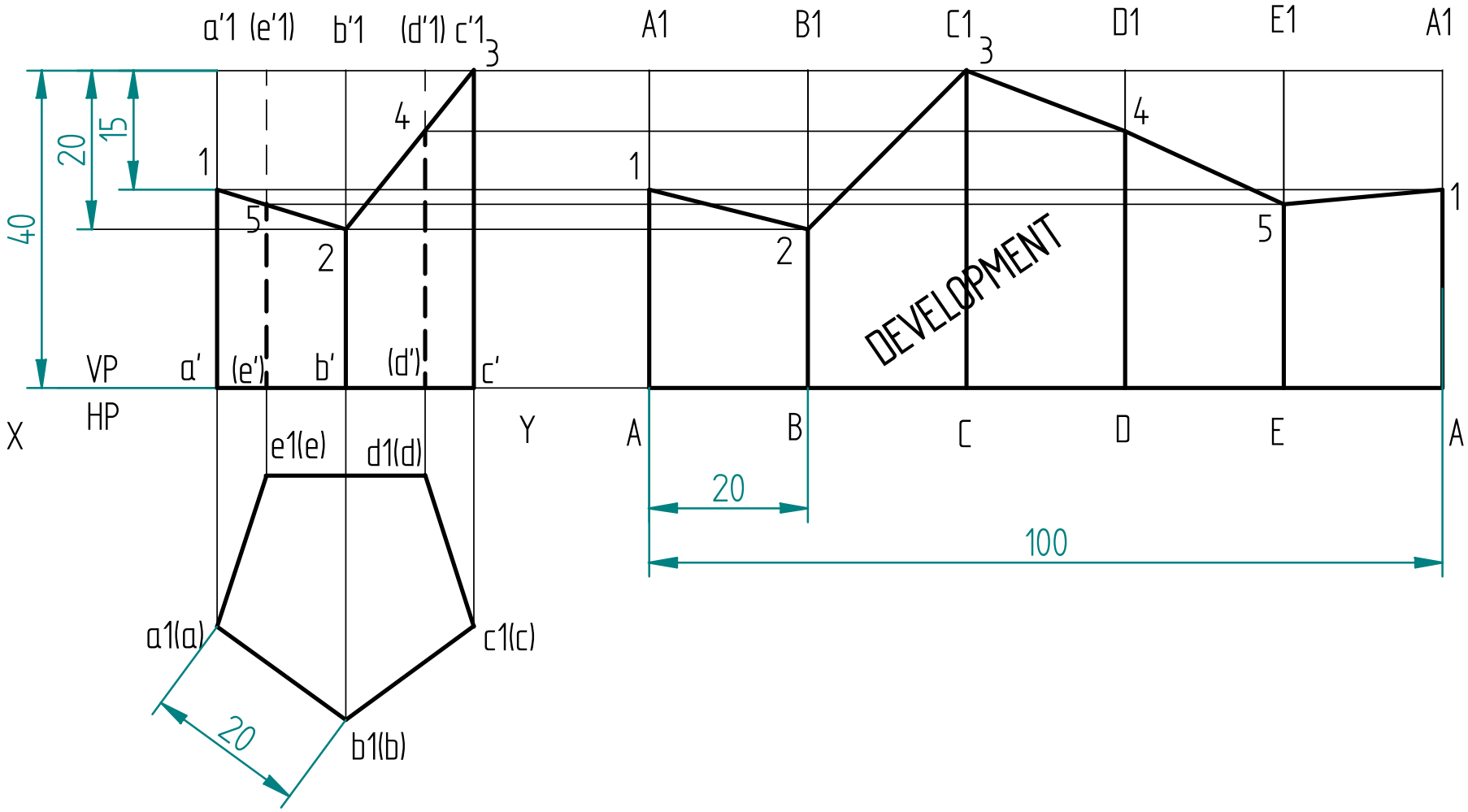
4.2. A square prism of 30mm side of the base and height 50mm is resting with its base on HP such that one of its vertical faces is inclined at 40^0 to VP.
It is cut as shown in the following front view figure. Draw the development of the lateral surface of the prism



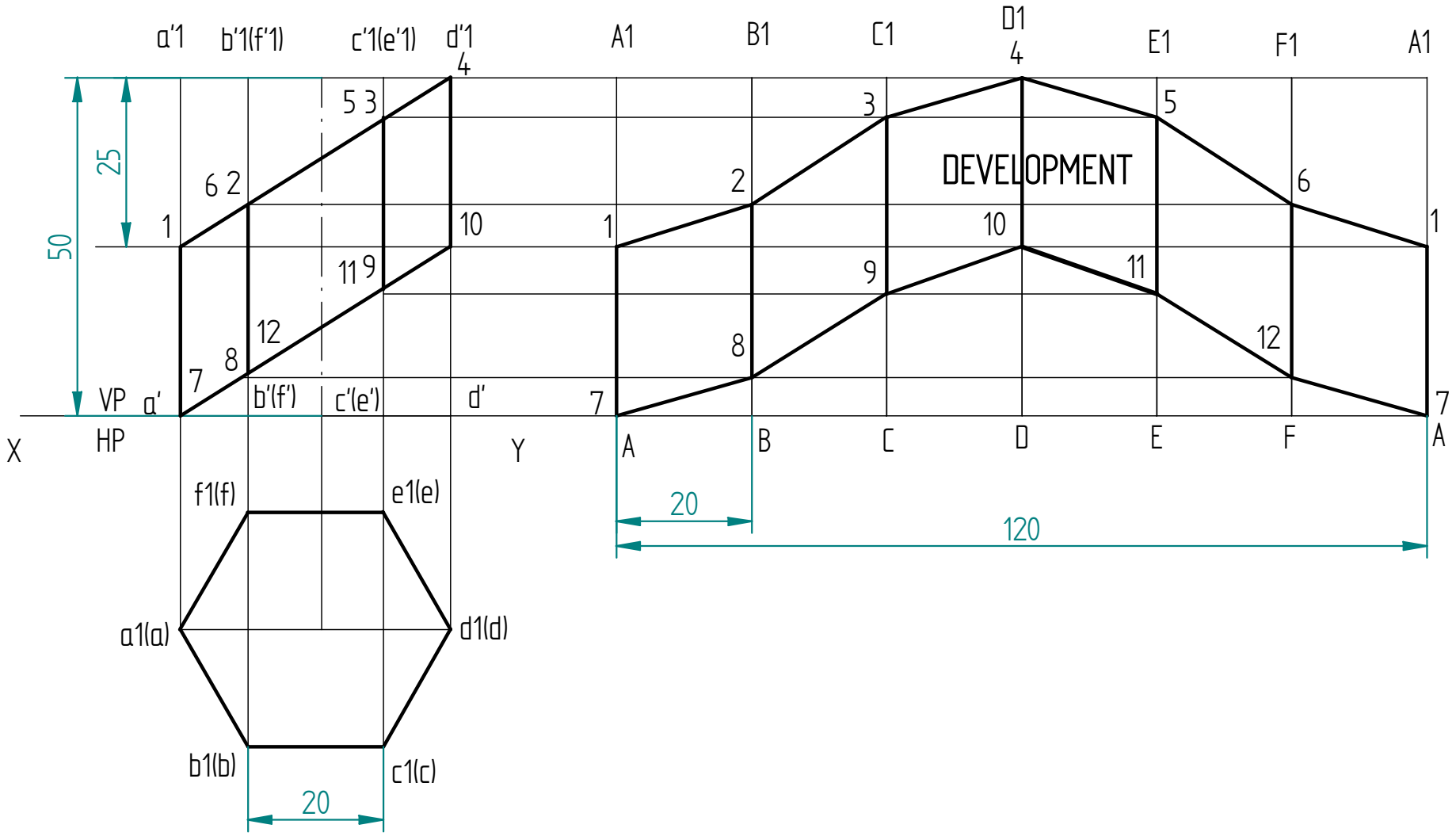
4.3 A rectangular prism of base 40mm × 25mm and height 65mm rests on HP on its base with the longer base side inclined at 30^0 to VP. It is cut by a plane inclined at 40^0 to HP; perpendicular to VP cuts the axis at its mid height. Draw the development of the remaining portion of the prism.



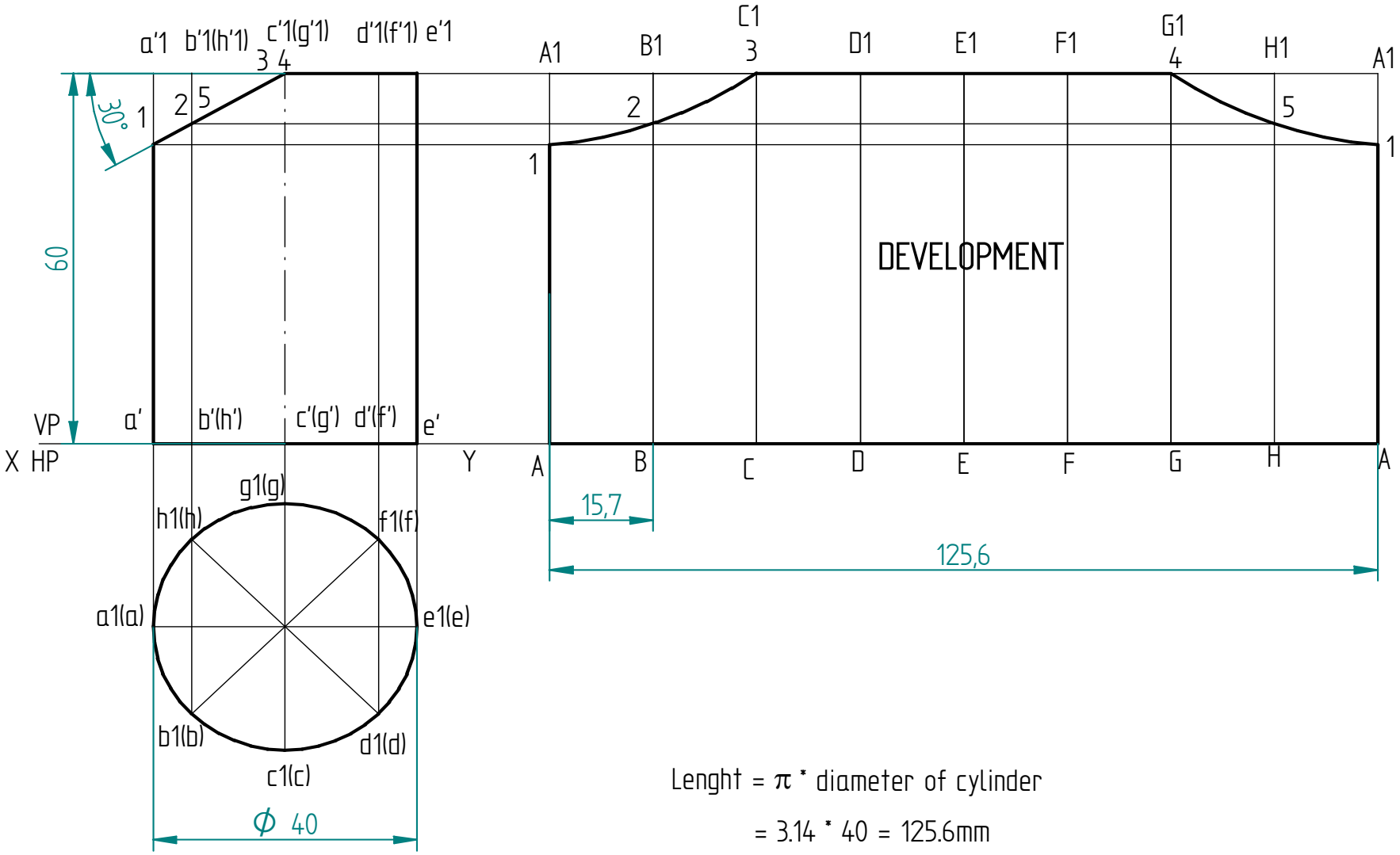
4.4 A pentagonal prism of base sides 20 mm and height 40 mm is resting with its base on HP with a base edge parallel to VP. The prism is cut as shown in the following front view. Draw the development of lateral surface of prism.



4.5 A hexagonal prism of base side 20mm and height 50mm is resting on HP on its base, such that one of its base edge is parallel to VP. The prism is cut in this position as shown in following front view. Draw the development of lateral surface of prism.



4.6 Develop the lateral surface of the cylinder of 40mm diameter and height 60mm which is cut in the following way.

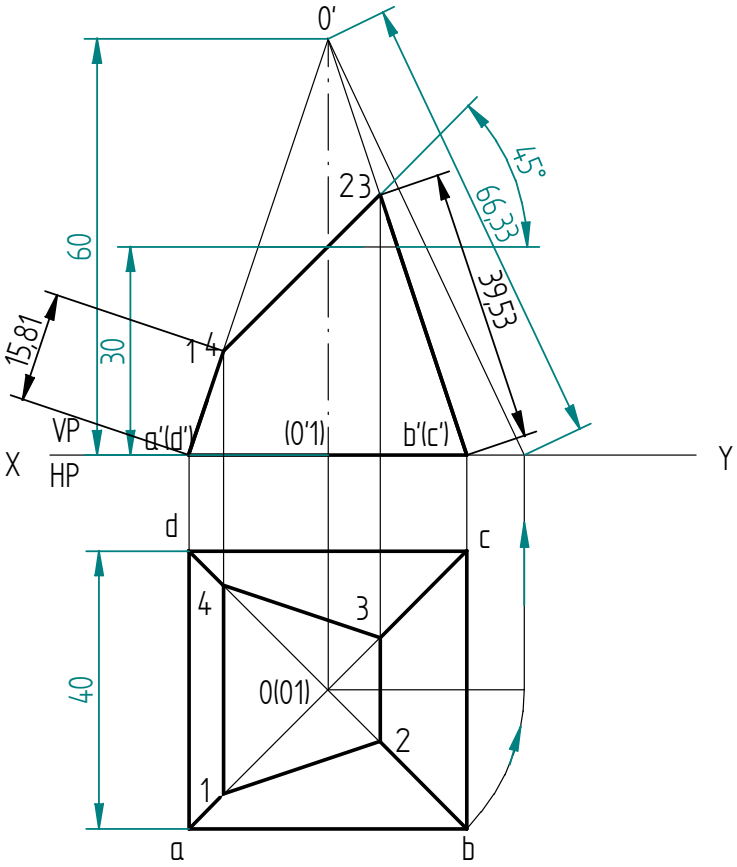


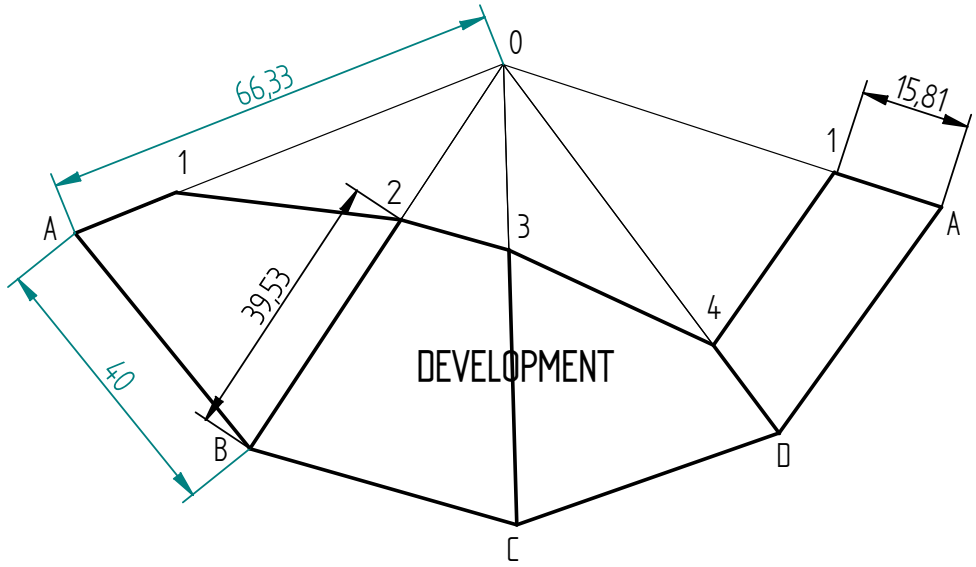
Lenght = π * diameter of cylinder

= 3.14 * 40 = 125.6mm

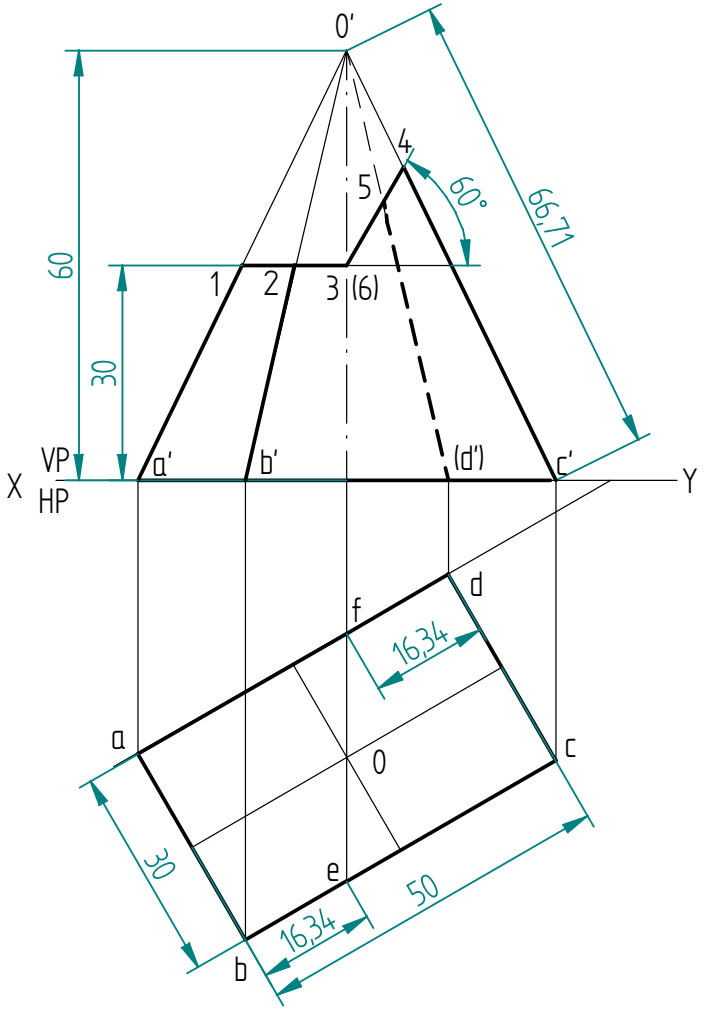
= 125.6 / 8 = 15.7mm

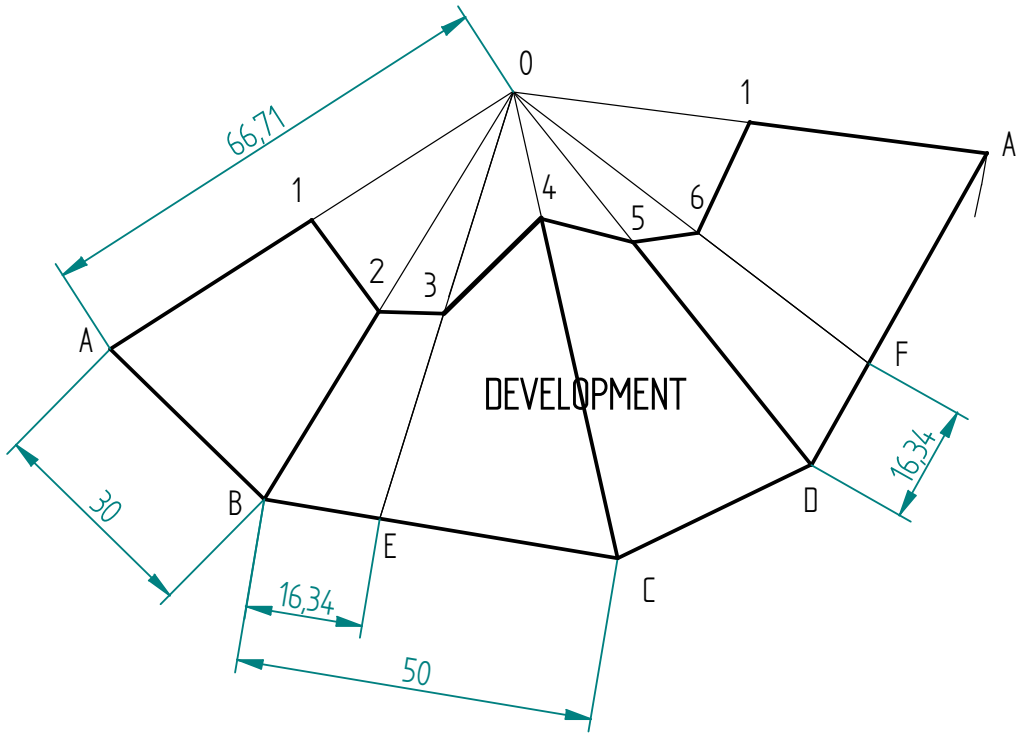
4.7 A square pyramid of base edge 40 mm and height 60 mm rests on HP with its axis vertical and two of its base edges parallel to VP. A section plane perpendicular to VP and inclined at 45^0 to HP bisects the axis of the pyramid. Draw the development of lateral surface of retained portion of the solid.



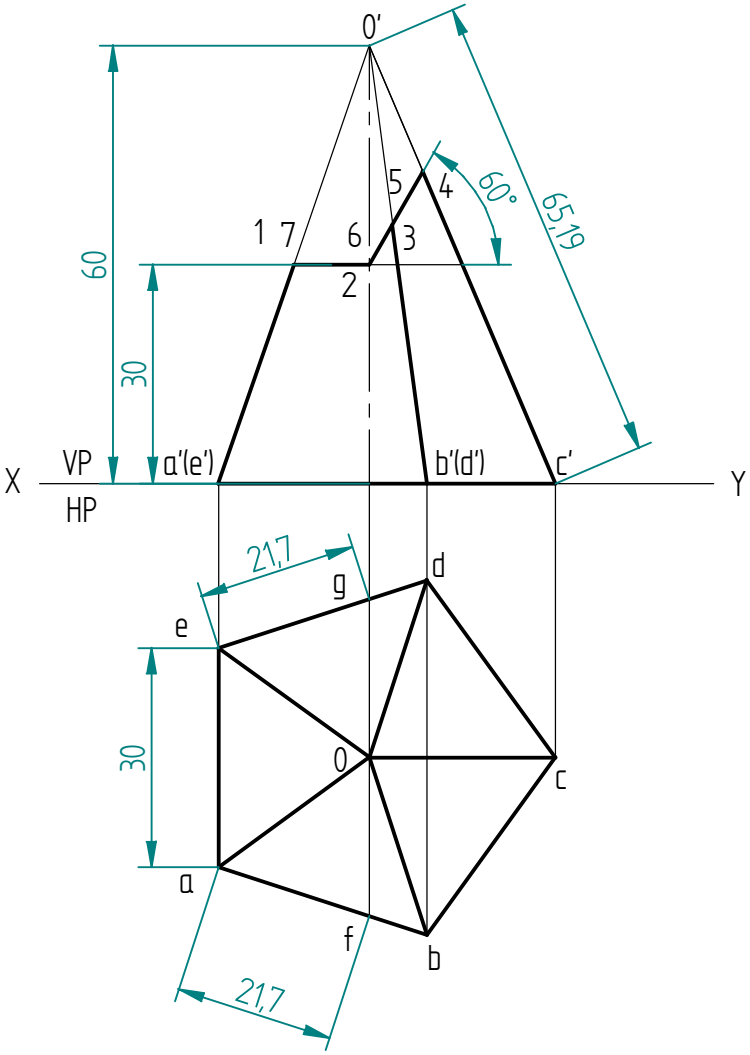


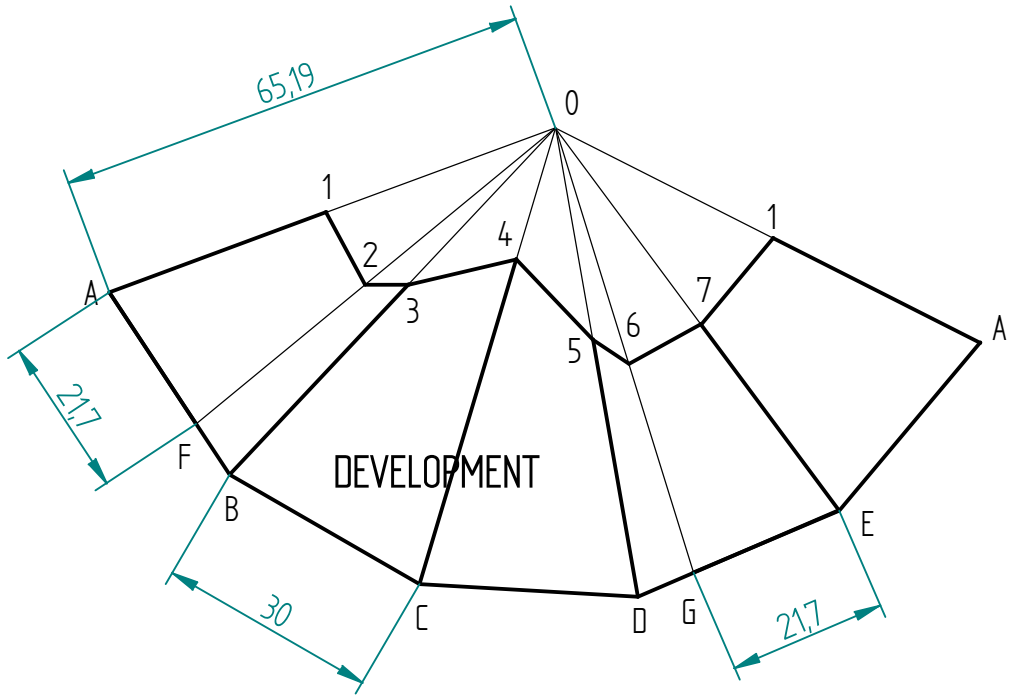
4.8 Draw the development of the lateral surface of the rectangular pyramid cut as shown in fig.



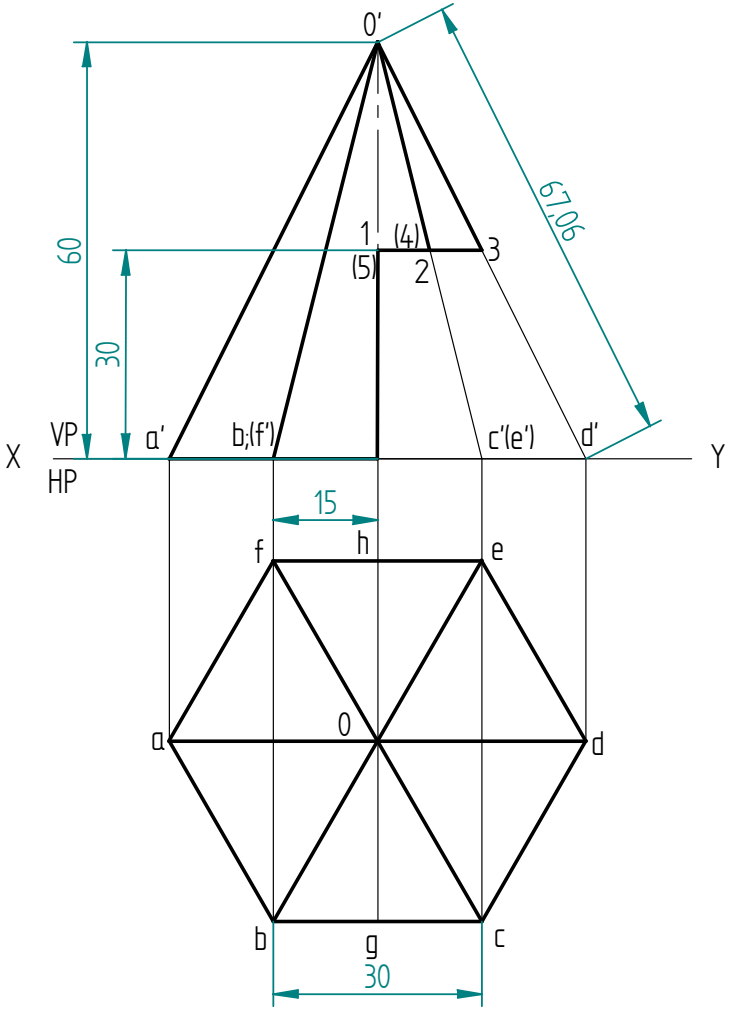


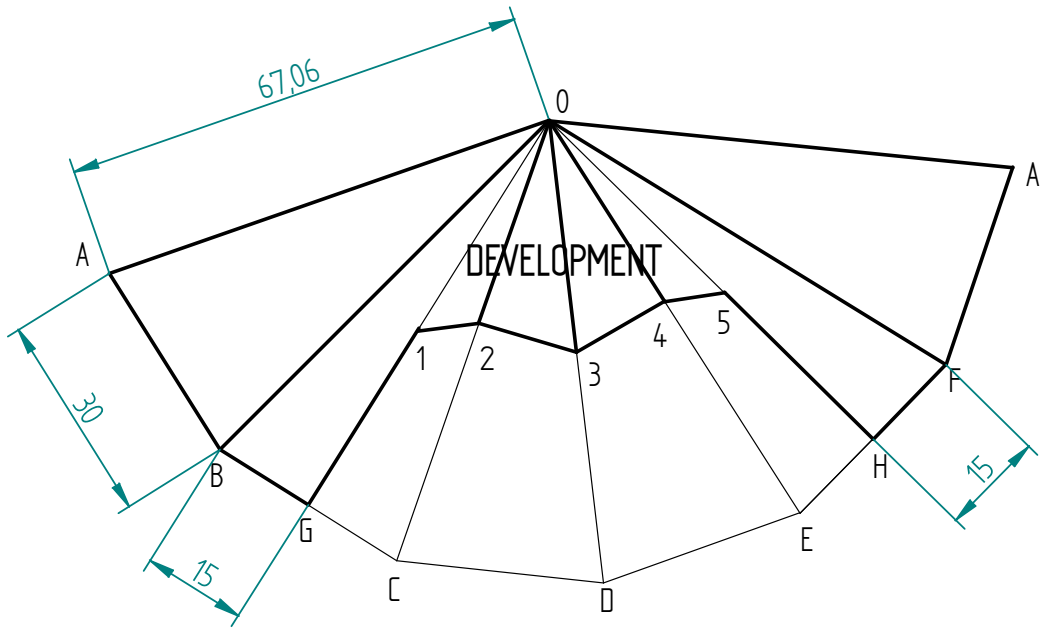
4.9 A pentagonal pyramid, 30mm sides, with a side of base perpendicular to VP. Draw the development of the lateral surfaces of the retained portion of the pyramid shown by the dark lines in the following figure.



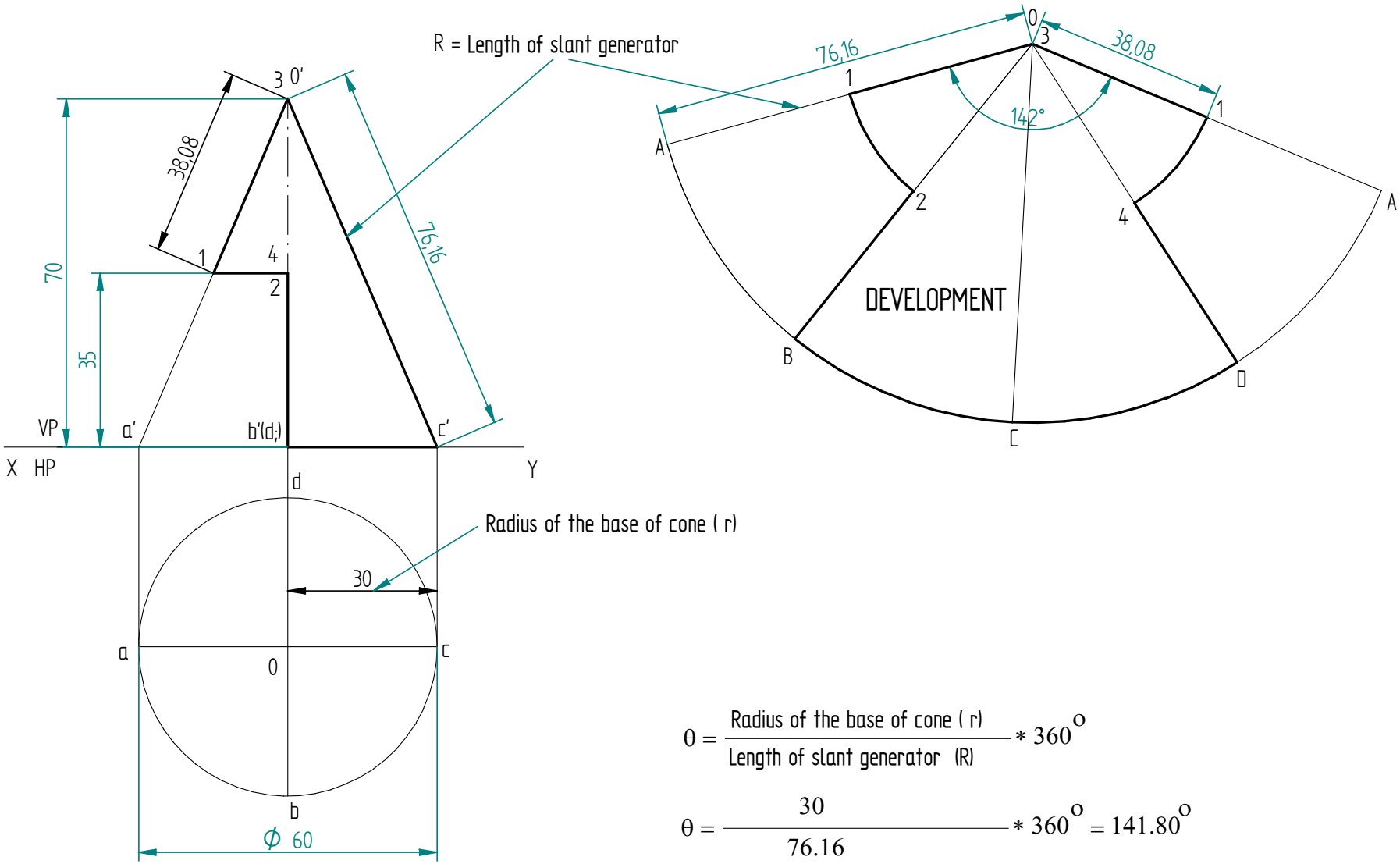


4.10 Draw the development of the lateral surface of the hexagonal pyramid shown in fig.





4.11 A cone of base diameter 60mm and height 70mm is resting on its base on HP. It is cut as shown in the following figure. Draw the development of the lateral surface of the remaining portion of the cone.

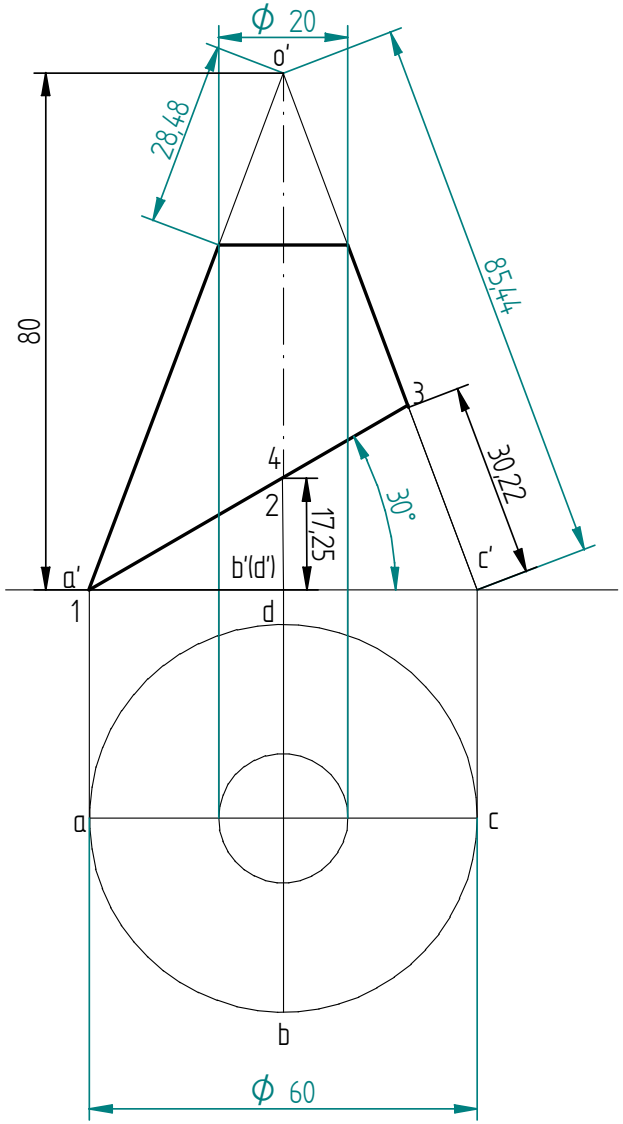


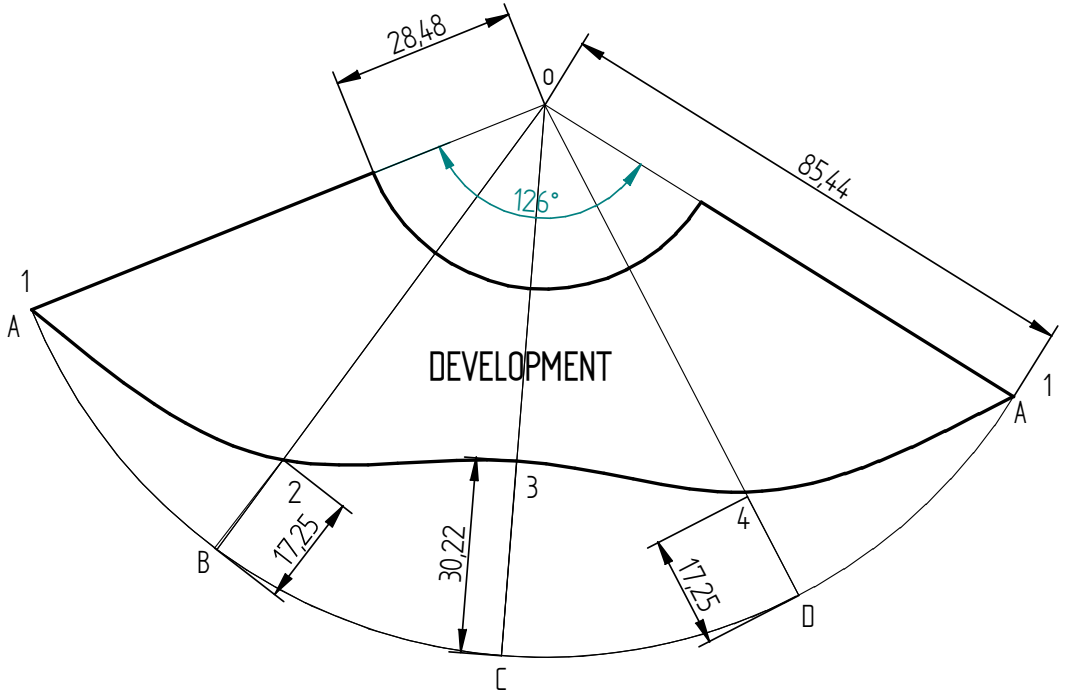
$$\theta = \frac{\text{Radius of the base of cone (r)}}{\text{Length of slant generator (R)}} * 360^{o}$$

$$\theta = \frac{30}{76.16} * 360^{o} = 141.80^{o}$$

$$\theta = \frac{141.80^{o}}{4} = 35.45^{o}$$

4.12 Draw the development of the lateral surface of the cone, whose front view is as shown in following figure



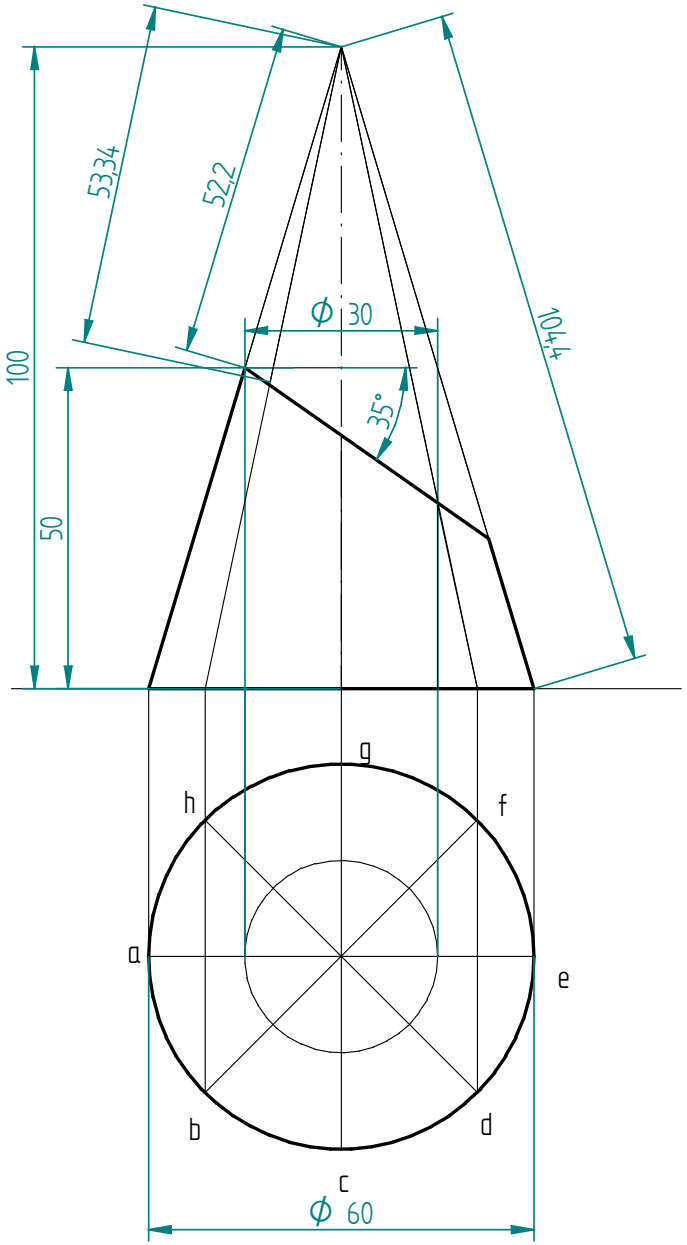


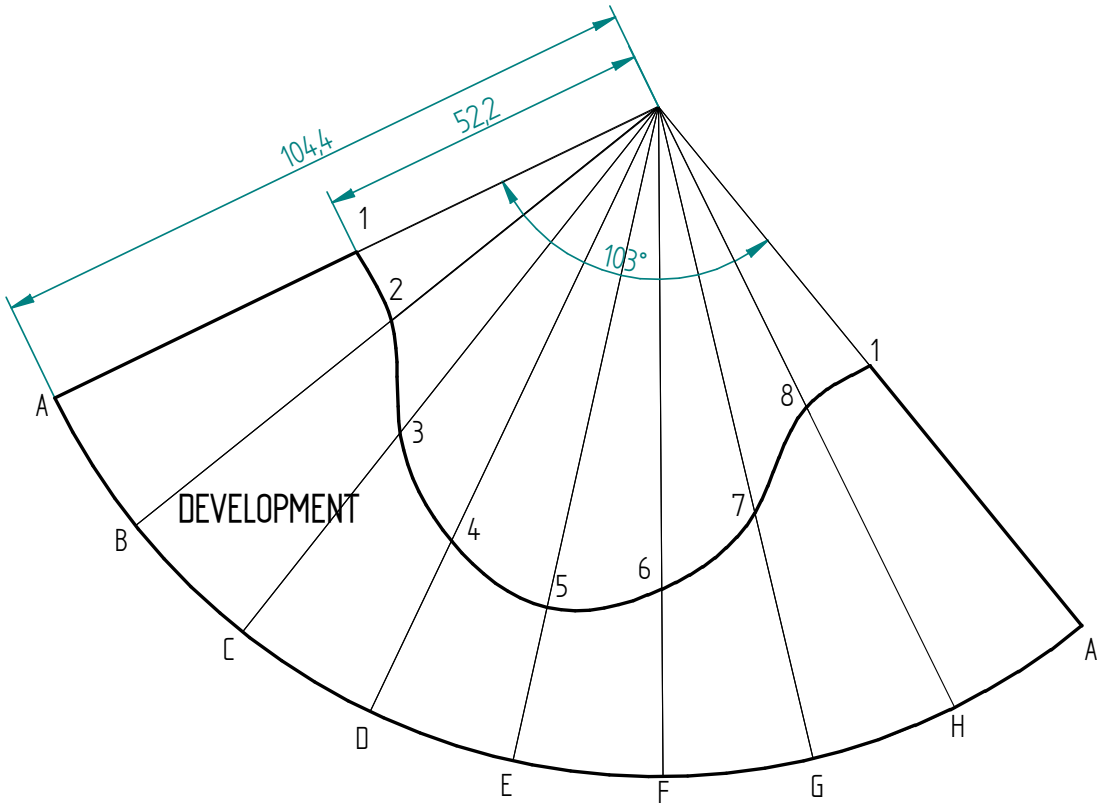
$$\theta = \frac{\text{Radius of the base of cone (r)}}{\text{Length of slant generator (R)}} * 360^{o}$$

$$\theta = \frac{30}{85.44} * 360^{o} = 126.40^{o}$$

$$\theta = \frac{126.40^{o}}{4} = 31..6^{o}$$

4.13 Draw the development of the lateral surface of the cone, whose front view is as shown in following figure



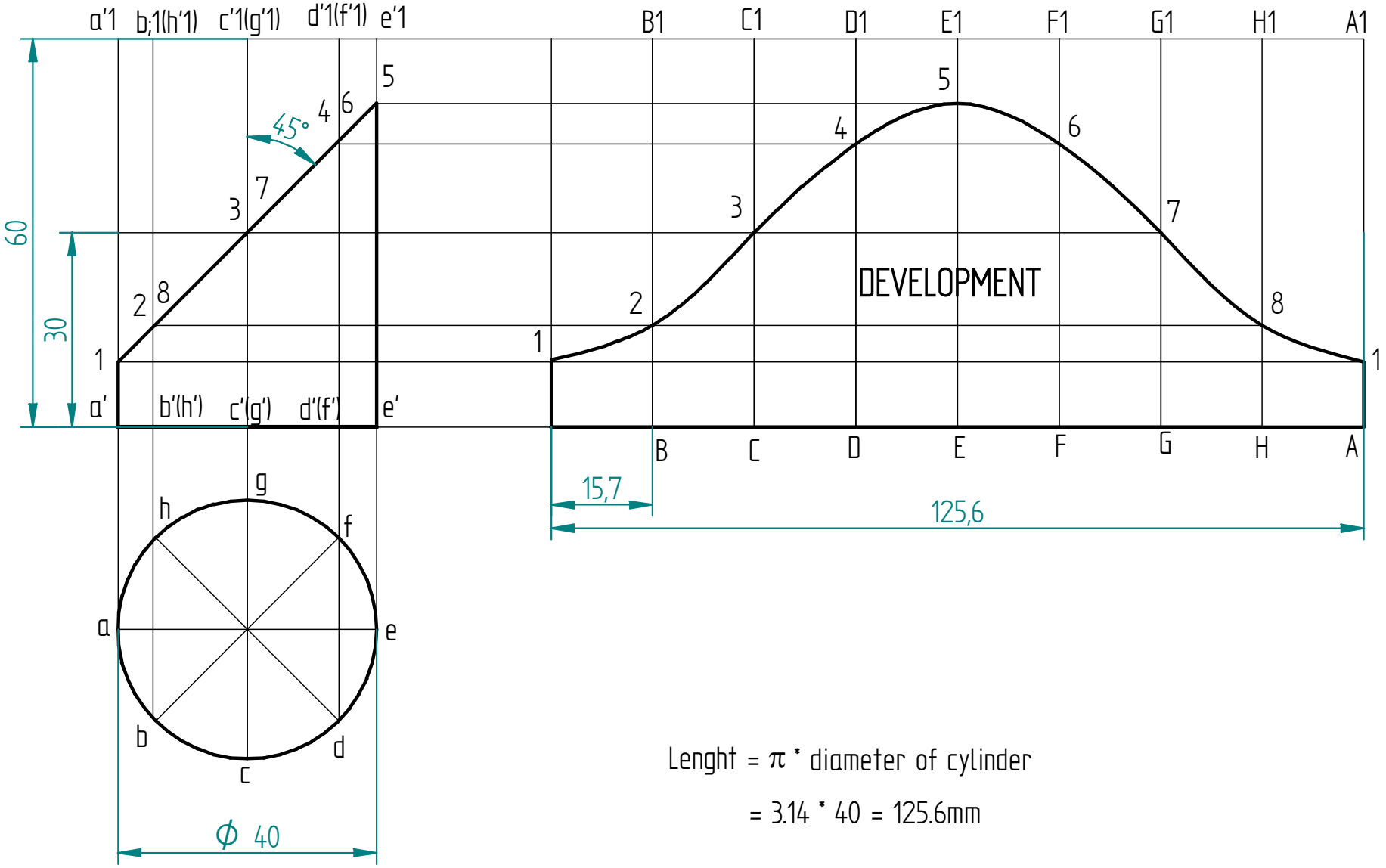


$$\theta = \frac{\text{Radius of the base of cone (r)}}{\text{Length of slant generator (R)}} * 360^{o}$$

$$\theta = \frac{30}{104.4} * 360^{o} = 103.44^{o}$$

$$\theta = \frac{103.44^{o}}{8} = 12.93^{o}$$

4.14. Develop the lateral surface of the cylinder of 40mm diameter and height 60mm which is cut in the following way.

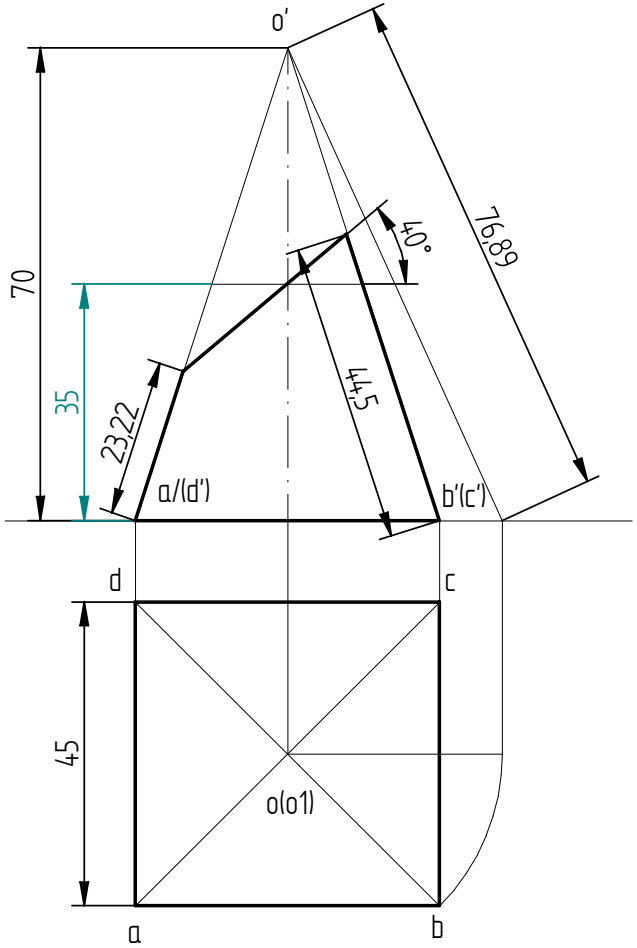


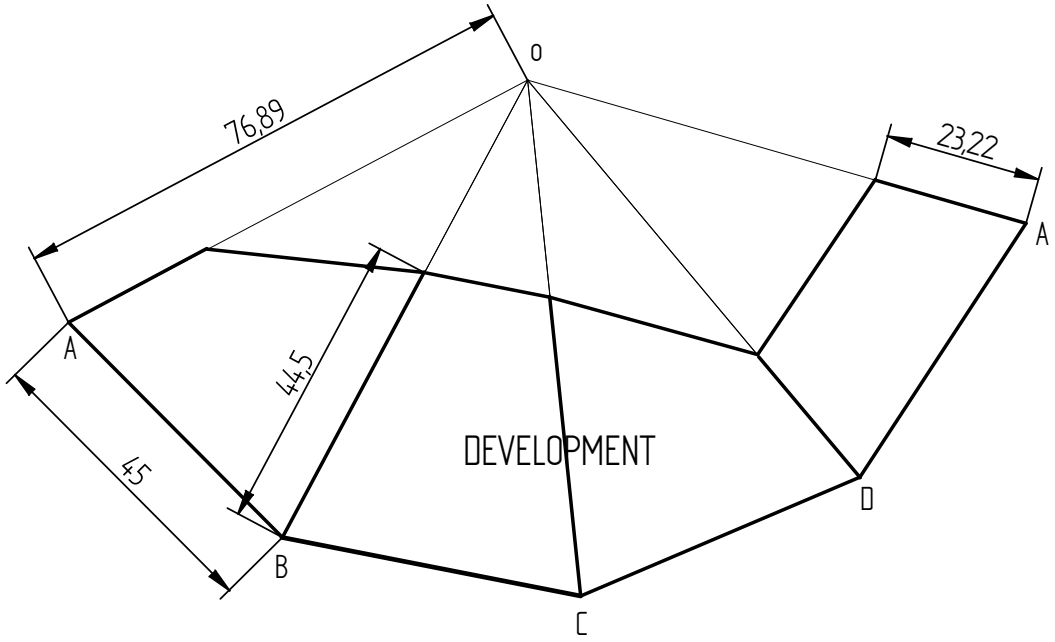
Lenght = π * diameter of cylinder

= 3.14 * 40 = 125.6mm

= 125.6 / 8 = 15.7mm

4.15. Draw the development of the lateral surface of the Square pyramid shown in fig.





4.16. Draw the development of the lateral surface of the hexagonal prism shown in fig.

